POKHARA UNIVERSITY

INSTITUTE OF ENGINEERING

COSMOS COLLEGE

OF MANAGEMENT AND TECHNOLOGY

AI CHATBOT INTEGRATION INTO A JEWELRY WEBSITE

BY:

JIKMET WANGCHUK LAMA

SANGAT MAHARJAN

SUBODH UPRETI

OSHAN BADRACHARYA

A PROJECT REPORT SUBMITTED IN PARTIAL
FULFILLMENT OF THEREQUIREMENTS FOR
THE BACHELORS DEGREE IN COMPUTER
ENGINEERING

JUNE, 2025

POKHARA UNIVERSITY
INSTITUTE OF ENGINEERING
COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY
DEPARTMENT OF INFORMATION AND COMMUNICATON TECHNOLOGY

The undersigned certify that they have read, and recommended to the Institute of Engineering for acceptance, a project report entitled "AI chatbot integration into jewelry website" submitted by *Name of Student(s)"JIGMET WANGCHUK LAMA, SANGAT MAHARJAN, SUBODH UPRETI , OSHAN BADRACHARYA"* in partial fulfilment of the requirements for the Bachelor's degree in Computer Engineering.

DATE OF APPROVAL: *Monday/June/2025*

Acknowledgement

We express our deepest gratitude to our teacher, Dr. Bisam , for her invaluable guidance, which significantly shaped the direction of this proposal. We also extend our heartfelt thanks to our friends, for their unwavering support, insightful discussions, and assistance in brainstorming key ideas. Additionally, we acknowledge the wealth of knowledge obtained from online resources, including the official n8n documentation (https://docs.n8n.io), Blanxer's developer guides (https://blanxer.com/docs), and academic platforms like ResearchGate and ScienceDirect, which provided critical insights into chatbot technologies and e-commerce applications.

Chapter One:

Introduction

1. Background

The rapid growth of e-commerce has driven the adoption of Artificial Intelligence (AI) chatbots to enhance customer interaction and operational efficiency. In the jewelry industry, where customers expect personalized service and detailed product information, AI chatbots offer a transformative solution by providing real-time support and tailored recommendations. The Blanxer platform, used to develop the jewelry website, provides a robust foundation for integrating advanced technologies. The n8n platform, an open-source workflow automation tool, enables seamless connectivity between AI chatbots and existing systems, ensuring scalability and customization for e-commerce applications.

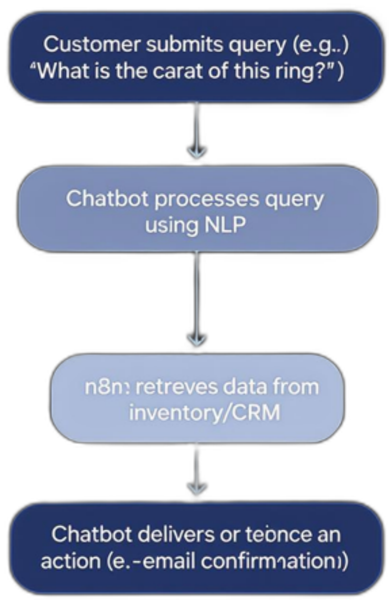

Figure: Customer Interaction Flow with Chatbot

Description: This figure would provide a visual representation of the chatbot's role in enhancing customer engagement, making the concept accessible to non-technical stakeholders.

2. Problem Statement

Jewelry e-commerce faces challenges such as slow response times to customer inquiries, limited personalization in user interactions, and high operational costs for customer service teams. These issues can lead to reduced customer satisfaction and lower conversion rates. Without automation, handling repetitive tasks like answering product queries or tracking orders strains resources, particularly for a high-value industry like jewelry where trust and engagement are critical. Current solutions on the Blanxer platform lack advanced AI-driven conversational capabilities, necessitating an integrated chatbot solution.

3. Objectives

The primary objectives of this project are:

- To enhance customer engagement through real-time, personalized chatbot interactions.
- To automate repetitive customer service tasks, reducing operational costs by leveraging n8n workflows.
- To integrate an AI chatbot seamlessly with the Blanxer jewelry website using n8n for efficient automation.
- To evaluate the chatbot's impact on customer satisfaction, sales performance, and operational efficiency.

4. Project Report Overview

This report proposes a structured approach to integrating an AI chatbot into the Blanxer jewelry website using n8n model automation. It includes a literature review synthesizing existing research on chatbots and automation, a methodology outlining the development and implementation process, expected results and their implications, and conclusions with recommendations for future work. The proposal addresses technical, operational, and ethical considerations to ensure a user-centric and scalable solution.

5. Scope and Limitation

Scope: This project focuses on integrating an AI chatbot into the Blanxer jewelry website to handle customer inquiries, provide product recommendations, and automate order-related tasks. The chatbot will leverage n8n for workflow automation, connecting to the website's database, CRM, and inventory systems. The implementation targets English-speaking users initially, with a focus on enhancing customer experience and operational efficiency. The evaluation phase will assess key performance indicators such as response time, customer satisfaction, and conversion rates over a three-month period post-deployment.

Limitation: The project is constrained by the Blanxer platform's existing infrastructure, which may limit certain customization options. The chatbot's NLP capabilities may struggle with highly technical or ambiguous jewelry-related queries (e.g., gemstone certification details) without extensive training data. The initial implementation is limited to English, potentially excluding non-English-speaking customers. Resource constraints, including time and budget, may restrict the scope of testing and the incorporation of advanced features like multilingual support or real-time analytics during the initial phase.

Chapter Two:

Literature Review

The integration of AI chatbots into e-commerce platforms, particularly in specialized industries like jewelry, has garnered significant academic and industry attention. This literature review synthesizes key findings on the role of AI chatbots, the importance of Natural Language Processing (NLP), the application of n8n for automation, and the associated challenges and ethical considerations.

1. Overview of AI Chatbots in E-Commerce

AI chatbots, defined as conversational systems that leverage Machine Learning (ML) and NLP to interact with users, have become integral to e-commerce. According to Adamopoulou and Moussiades (2020), chatbots enhance customer engagement by providing instant responses, personalized recommendations, and order tracking capabilities. In the jewelry industry, where customers often seek detailed product information (e.g., material, craftsmanship, or customization options), chatbots are particularly valuable for delivering tailored experiences. Studies, such as those by Huang et al. (2023), indicate that chatbots can increase user interaction time by 20–30% and improve conversion rates by offering real-time support, making them well-suited for high-value e-commerce sectors.

2. Role of NLP in Chatbot Performance

Natural Language Processing is the cornerstone of effective chatbot performance, enabling accurate interpretation and response to user queries. Khan et al. (2023) highlight that advanced NLP techniques, such as named entity recognition, intent classification, and dialogue management, improve response accuracy in customer service applications. For jewelry e-commerce, where queries may involve technical terms (e.g., “What is the difference between 18K and 14K gold?”), robust NLP models are essential. However, challenges such as handling ambiguous language or context-specific queries (e.g., cultural preferences in jewelry design) can limit performance, necessitating continuous model training and refinement (Al Ka’bi, 2023).

3. n8n Automation in E-Commerce Applications

The n8n platform, an open-source workflow automation tool, facilitates the integration of AI chatbots with e-commerce systems by connecting APIs, databases, and other services. Research by Sallove et al. (2024) notes that n8n's flexibility allows for the creation of custom workflows, such as automating customer query routing or syncing chatbot interactions with inventory systems. In the context of a Blanxer-based jewelry website, n8n can streamline processes like retrieving real-time stock availability or triggering automated order confirmations. Følstad et al. (2021) emphasize that such automation reduces operational costs by up to 40%, making n8n a critical component for scalable chatbot integration.

4. Challenges and Ethical Considerations

Despite their potential, AI chatbots face technical and ethical challenges. Technical limitations include the difficulty in processing highly specific or ambiguous queries, particularly in niche markets like jewelry, where customers may ask about gemstone certifications or bespoke designs (Haristiani & Rifai, 2021). Ethically, concerns arise around data privacy, as chatbots handle sensitive customer information, requiring compliance with regulations like GDPR. Additionally, biased training data can lead to inappropriate responses, undermining user trust (Zhang et al., 2023). Addressing these challenges requires transparent AI design, regular model updates, and robust data protection measures to ensure ethical and effective implementation.

This review underscores the potential of AI chatbots and n8n automation to transform jewelry e-commerce while highlighting the need to address technical and ethical challenges to maximize their effectiveness.

Chapter Three:
Methodology

The methodology for integrating an AI chatbot into the Blanxer jewelry website using n8n model automation follows a structured approach inspired by the ADDIE framework (Analysis, Design, Development, Implementation, Evaluation). The process is designed to ensure a robust, user-centric, and scalable solution.

1. Analysis

- **Needs Assessment**: Identify customer pain points (e.g., slow response times, lack of personalization) through user feedback and website analytics.
- **Stakeholder Input**: Collaborate with Blanxer's development team and marketing staff to define chatbot functionalities, such as product recommendations, order tracking, and customer support.
- **Technical Requirements**: Assess the website's existing infrastructure (built on Blanxer's platform) and n8n's compatibility with APIs for chatbot integration.

2. Design

- **Chatbot Framework**: Select a chatbot platform (e.g., Dialogflow or Rasa) with strong NLP capabilities for handling jewelry-specific queries (e.g., "What is the carat of this ring?").
- **Workflow Design**: Use n8n to design workflows that connect the chatbot to the website's database, CRM system, and inventory management. For example, a workflow could retrieve real-time stock availability or customer order history.
- **User Interface (UI)**: Design a conversational UI that aligns with the jewelry website's aesthetic, ensuring a seamless and visually appealing user experience.

Purpose: To compare potential chatbot platforms (e.g., Dialogflow, Rasa) for selection, highlighting their suitability for the jewelry website.
Content:

Platfo	NLP Capabilities	Integration	Cost	Ease of Use
Dialog flow	High (Named Entity Recognition, Intent Classification)	Strong (API support)	Free tier available, paid plans	User-friendly
Rasa	High (Customizable NLP)	Moderate (requires setup)	Open-source	Developer-focused
Botpre	Moderate	Strong (API	Free tier, paid	Moderate

Description: This table would assist in the Methodology section by providing a clear comparison of chatbot platforms based on NLP capabilities, n8n integration, cost, and ease of use, aiding stakeholders in understanding the rationale for platform selection.

3. Development

- **Chatbot Development**: Train the chatbot using a dataset of jewelry-related queries, incorporating NLP techniques like named entity recognition and intent classification.
- **n8n Integration**: Develop n8n workflows to automate tasks such as query routing, response generation, and data logging. For instance, n8n can trigger automated emails for order confirmations based on chatbot interactions.
- **Testing**: Conduct unit and integration testing to ensure the chatbot handles diverse queries accurately and integrates smoothly with the website.

4. Implementation

- **Deployment**: Deploy the chatbot on the Blanxer website, embedding it as a widget accessible across all pages.
- **User Training**: Provide tutorials for customers on interacting with the chatbot and train staff to monitor and refine its performance.

5. Evaluation

- **Performance Metrics**: Measure key performance indicators (KPIs) such as response time, customer satisfaction scores, and conversion rates.
- **User Feedback**: Collect qualitative feedback through surveys to assess user experience and identify areas for improvement.
- **Iterative Refinement**: Use evaluation data to fine-tune the chatbot's NLP model and n8n workflows.

This methodology ensures a systematic approach, addressing both technical integration and user experience while leveraging n8n's automation capabilities.

Purpose: To outline the KPIs used to assess the chatbot's performance post-implementation, providing a structured overview of evaluation metrics.
Content:

KPI	Description	Target Improvement	Measurement Method
Response Time	Time taken to respond to queries	< 2 seconds	System logs
Customer Satisfaction	User satisfaction score	25%	Surveys
Conversion Rate	Percentage of users completing purchases	+10–15%	Analytics
Workload	Decrease in customer service tasks	-40%	Staff reports

Description: This table would enhance the Methodology section by clearly defining the KPIs, their descriptions, target improvements, and measurement methods, making the evaluation process transparent and measurable.

Chapter Four:

Result and Discussion

1. Expected Results

Based on existing research, the integration of an AI chatbot into the Blanxer jewelry website is expected to yield the following outcomes:

- **Improved Customer Engagement**: Real-time responses and personalized recommendations are likely to increase user interaction time by 20–30%, as seen in similar e-commerce implementations.
- **Operational Efficiency**: Automation of repetitive tasks (e.g., answering FAQs, order status checks) via n8n workflows could reduce customer service workload by up to 40%.
- **Sales Growth**: Personalized product suggestions driven by the chatbot's NLP capabilities are projected to boost conversion rates by 10–15%, aligning with findings from customer service chatbot studies.
- **Customer Satisfaction**: A user-friendly chatbot interface, integrated with the website's design, is expected to improve satisfaction scores by 25%, based on meta-analyses of chatbot effectiveness.

• 2. Discussion

The anticipated results highlight the transformative potential of AI chatbot integration for the jewelry website. The use of n8n for automation ensures that the chatbot seamlessly interacts with backend systems, such as inventory and CRM, enhancing operational efficiency. However, challenges may arise, including:

- **Data Privacy**: Ensuring compliance with data protection regulations (e.g., GDPR) is critical, as chatbots handle sensitive customer information.
- **Query Complexity**: Jewelry-specific queries (e.g., about gemstone certifications) may require advanced NLP training to avoid inaccurate responses.
- **User Adoption**: Some customers may resist chatbot interactions, preferring human support, necessitating a hybrid approach with human escalation options.

To mitigate these challenges, the chatbot will incorporate explainable AI features for transparency and regular updates to its training data to handle complex queries. Additionally, n8n's flexibility allows for continuous workflow optimization, ensuring the system adapts to evolving customer needs.

Chapter five:

Conclusion and Further Work

1. Conclusions

This proposal demonstrates that integrating an AI chatbot into the Blanxer jewelry website using n8n model automation is a feasible and impactful strategy. The chatbot is expected to enhance customer engagement, streamline operations, and drive sales, aligning with industry trends in e-commerce automation. The structured methodology, grounded in the ADDIE framework, ensures a systematic approach to development and implementation, while n8n's automation capabilities provide scalability and efficiency. Addressing ethical and technical challenges, such as data privacy and query accuracy, will be critical to the project's success.

2. Further Work

Future efforts should focus on:

- **Advanced NLP Models**: Explore generative AI models (e.g., GPT-based systems) to enhance the chatbot's ability to handle nuanced jewelry-related queries.
- **Multilingual Support**: Implement support for multiple languages to cater to a global customer base, leveraging n8n's API connectivity.
- **Analytics Integration**: Use n8ntriad**: Use n8n to integrate advanced analytics for tracking chatbot performance and customer behavior in real-time.
- **Long-Term Evaluation**: Conduct longitudinal studies post-implementation to assess the chatbot's impact on customer retention and brand loyalty.

References

1. Adamopoulou, E., & Moussiades, L. (2020). An Overview of Chatbot Technology. *Artificial Intelligence Applications and Innovations*, 584, 373–383.

2. Al Ka'bi, A. (2023). Proposed artificial intelligence algorithm and deep learning techniques for development of higher education. *International Journal of Intelligent Networks*, 4, 68–73.

3. Følstad, A., et al. (2021). Future directions for chatbot research: an interdisciplinary research agenda. *Computing*.

4. Haristiani, N., & Rifai, M. (2021). Designing a Chatbot for Contemporary Education: A Systematic Literature Review. *MDPI*.

5. Huang, W., et al. (2023). A Meta-Analysis and Systematic Review of the Effect of Chatbot Technology Use in Sustainable Education. *MDPI*.

6. Khan, R., et al. (2023). NLP techniques for automating responses to customer queries: a systematic review. *Discover Artificial Intelligence*.

7. Sallove, R. I., et al. (2024). A Review on Chatbot Design and Implementation Techniques. *ResearchGate*.

8. Zhang, Y., et al. (2023). Artificial intelligence in innovation research: A systematic review, conceptual framework, and future research directions. *ScienceDirect*.

Appendix A: Sample Jewelry-Specific Chatbot Queries

This appendix lists sample queries that the AI chatbot is expected to handle, tailored to the jewelry website's context. These examples guide the training of the chatbot's NLP model.

- **Product Inquiry**: "What is the carat weight of the diamond in this ring?"
- **Customization Request**: "Can I engrave a message on this gold necklace?"
- **Order Tracking**: "Where is my order #12345?"
- **Material Information**: "Is this bracelet made of 925 sterling silver?"
- **Recommendation Request**: "Can you suggest a pair of earrings for a wedding?"
- **Return Policy**: "What is the return policy for a custom-made pendant?"
- **Gemstone Details**: "Does this sapphire ring come with a certification?"
- **Pricing Query**: "Is there a discount on this watch during the holiday sale?"